

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	01 Dec 2015	First Release	ONG KEAN SHENG
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

Content

PSP 2 FRONT MODULE OFFLINE ADJUSTMENT.....	3
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	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

PSP 2 FRONT MODULE OFFLINE ADJUSTMENT

Steps:

1. Adjust the DIP switch at Front Projector Module as below Figure 1.

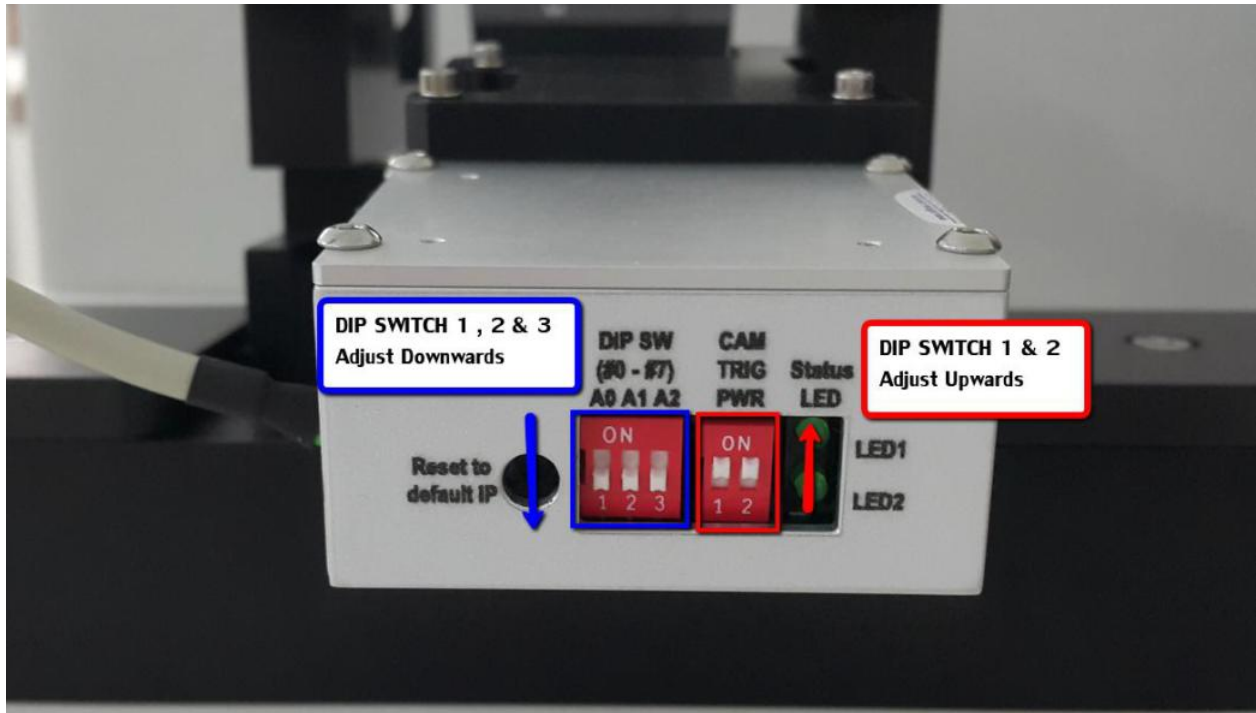


Figure 1 Front Projector IP DIP SWITCH CONFIGURATION

2. Assemble the PSP 2 Module at "S2EX PSP Adjustment Offline Jig" by using M5 X 12 as below Figure 2.

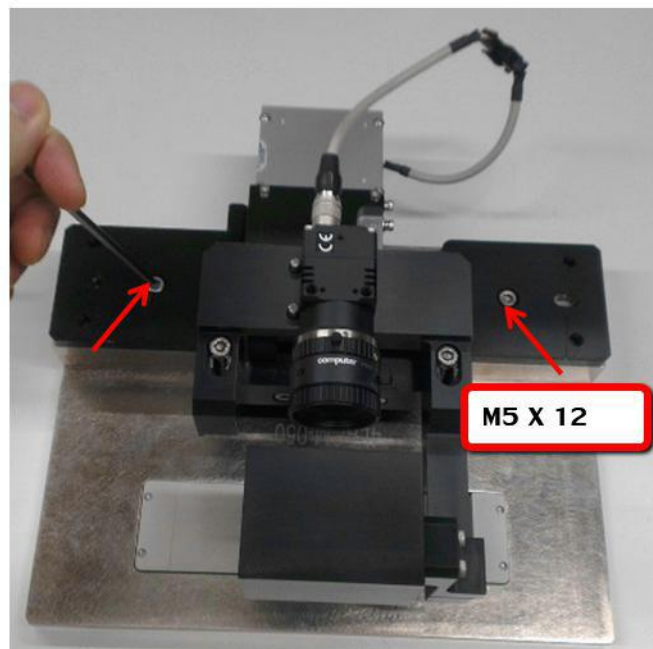


Figure 2 S2EX OFFLINE JIG ASSEMBLY

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

3. Open Front Access Door , gently allocate “S2EX PSP Adjustment Offline Jig ” (Part ID- 4XASPP-A003) at Granite area (May refer to below Figure 3 for the desire location)

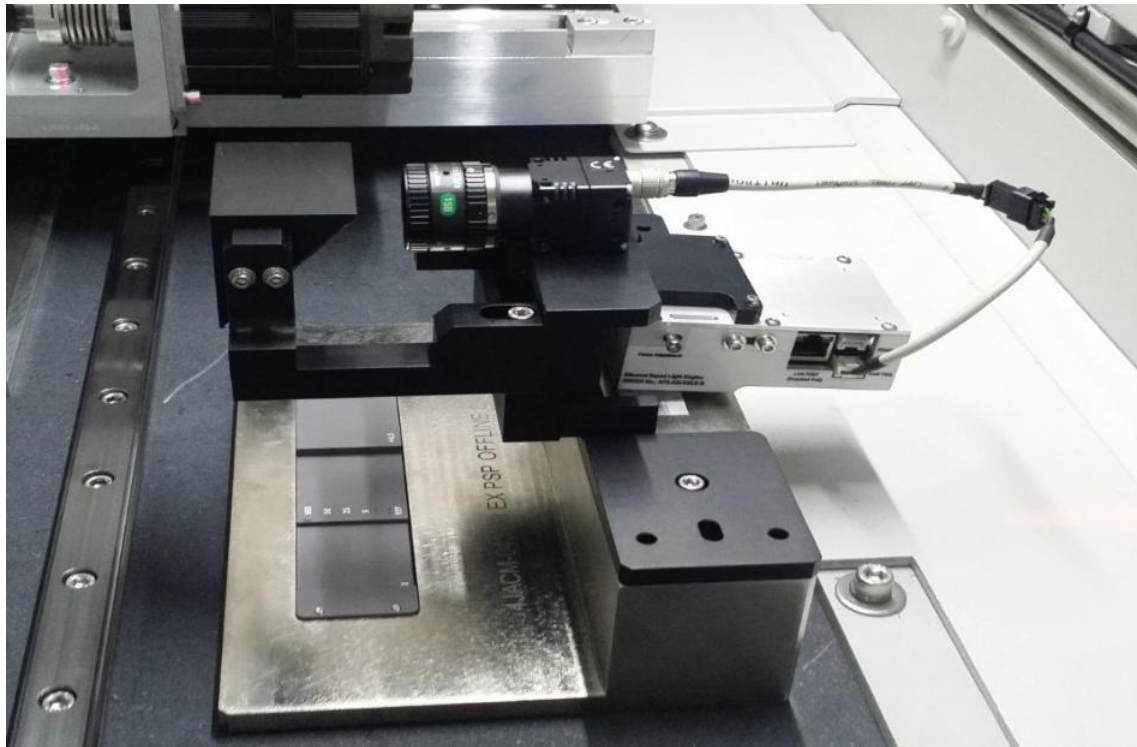


Figure 3 OFFLINE JIG & MODULE LOCATION

4. Turn Off 3 PIN PLUG Switch as below Figure 4. (Remarks : Power supply for POE Switch)

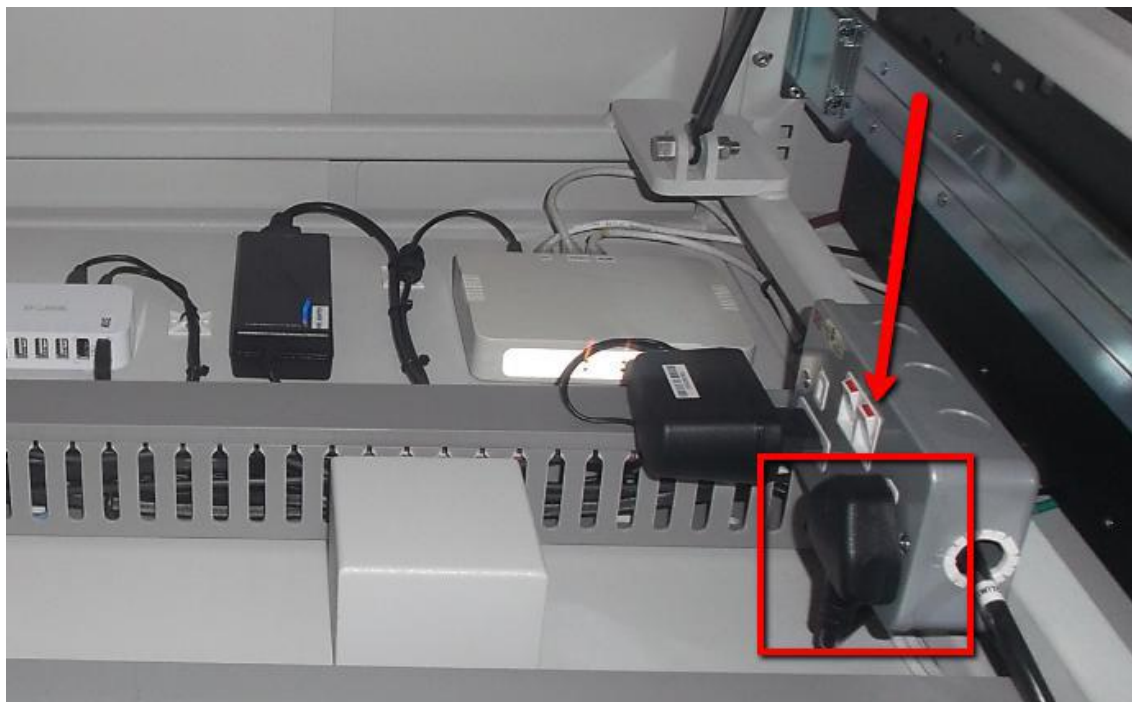


Figure 4 3 PIN PLUG POWER SWITCH

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

- Perform Marking at below cable and unplug Front Cat 5 cable at POE SWITCH (which allocated at rear server area) as below Figure 5.

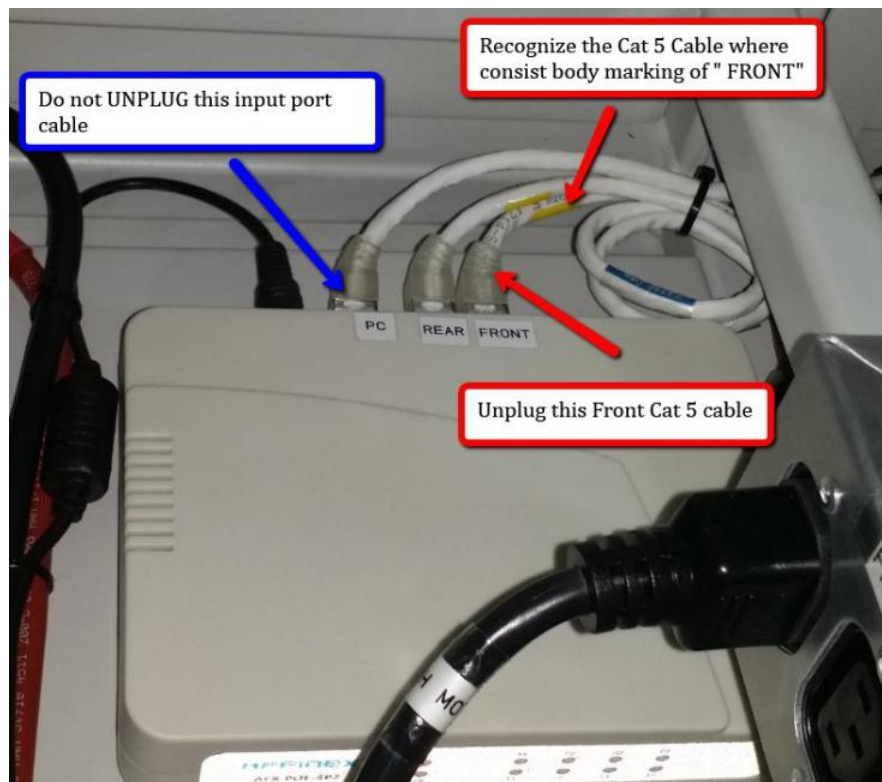


Figure 5 POE SWITCH CAT 5 CABLE

- Using another 1 x 2530-0063 (30 Feet Cat cable) and plug at POE Switch's Front Labeling Port" as below Figure 6.



Figure 6 Front labeling Cat 5 Port

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

7. The other end of Cat 5 cable will be connecting to PSP 2 Projector Port as below Figure 7.

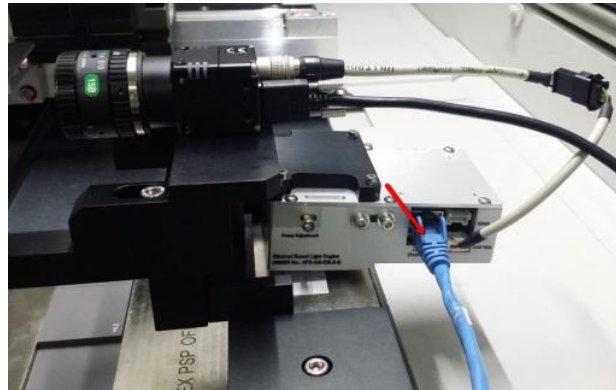


Figure 7 PSP 2 PROJECTOR PORT

8. Using 1 x 2530-0110 (Camera Cable_Area Scan_5Meter) and connect to PSP 2 Camera as below Figure 8.

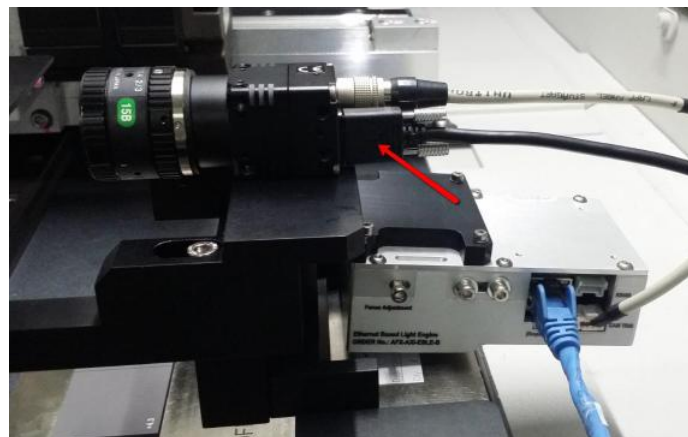


Figure 8 PSP 2 Camera Cable

Remarks: Due to the Camera Cable Length, rear door may require to open to prolong the cable routing to master controller. Please refer to Figure 9.

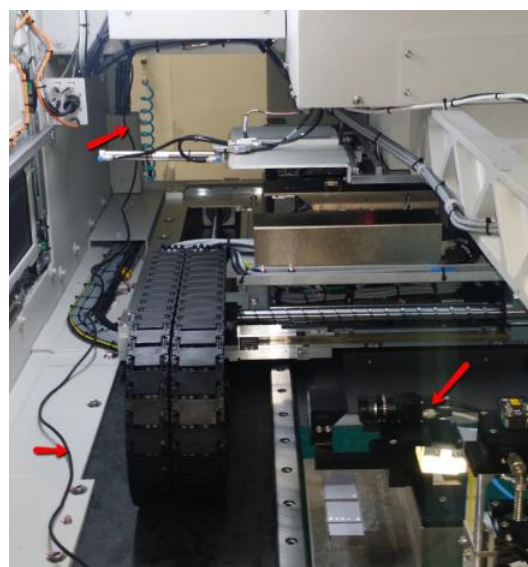


Figure 9 Camera Cable Routing (Reference Only)

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

9. The other end of the PSP 2 CAMERA CABLE may connect to Master Controller (USB PORT 3.0 Blue Color) as below Figure 10.

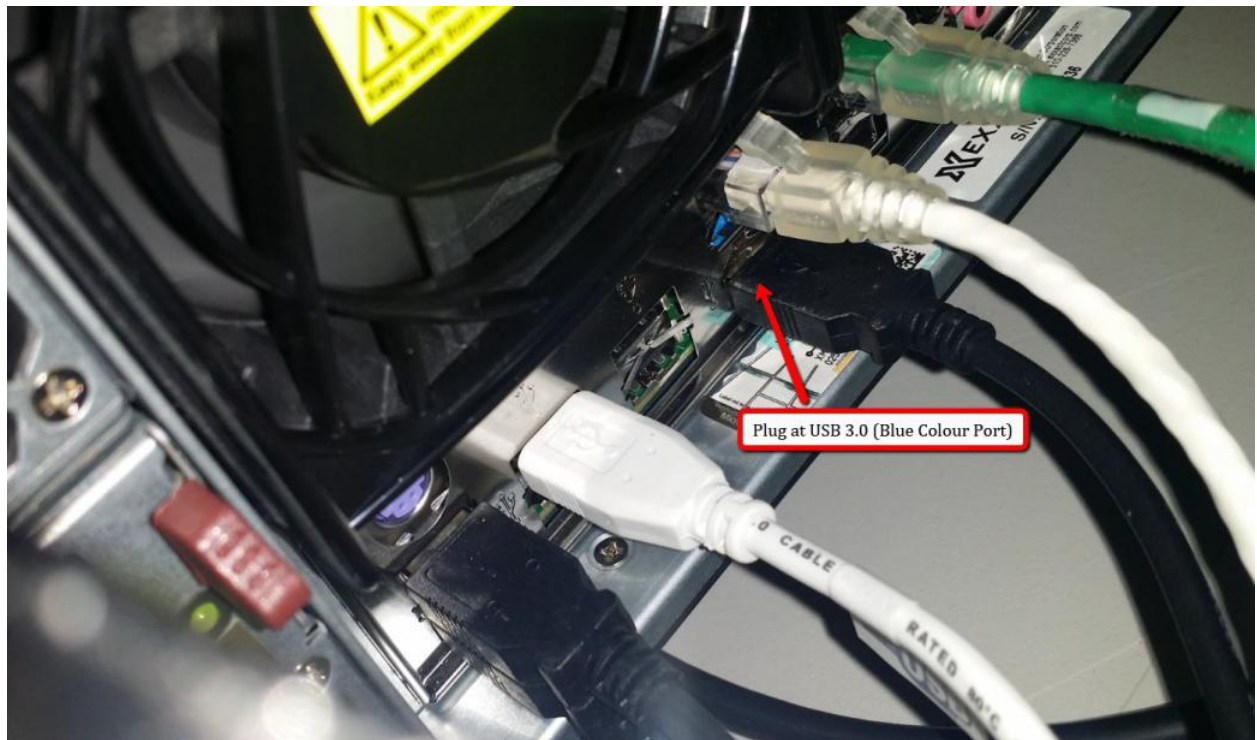


Figure 10 Rear side of Master controller (USB 3.0 Port)

10. Turn on 3 PIN PLUG Switch as below Figure 11.

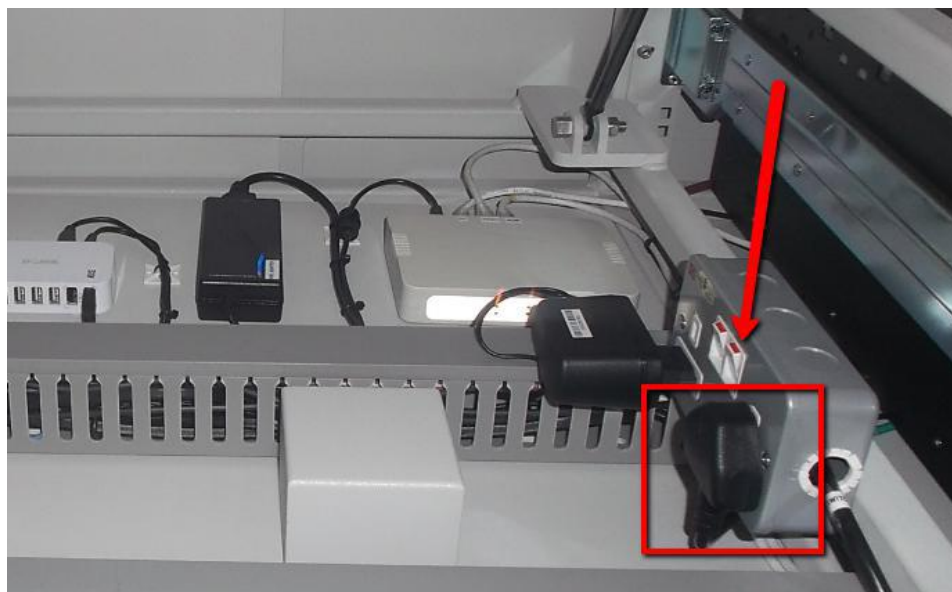


Figure 11 3PIN PLUG POWER SWITCH

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

11. Go to Control Panel -> Network and Internet -> View Network and tasks as below Figure 12.



Figure 12 Network and Internet

12. Choose "Change Adapter Settings" as below Figure 13.

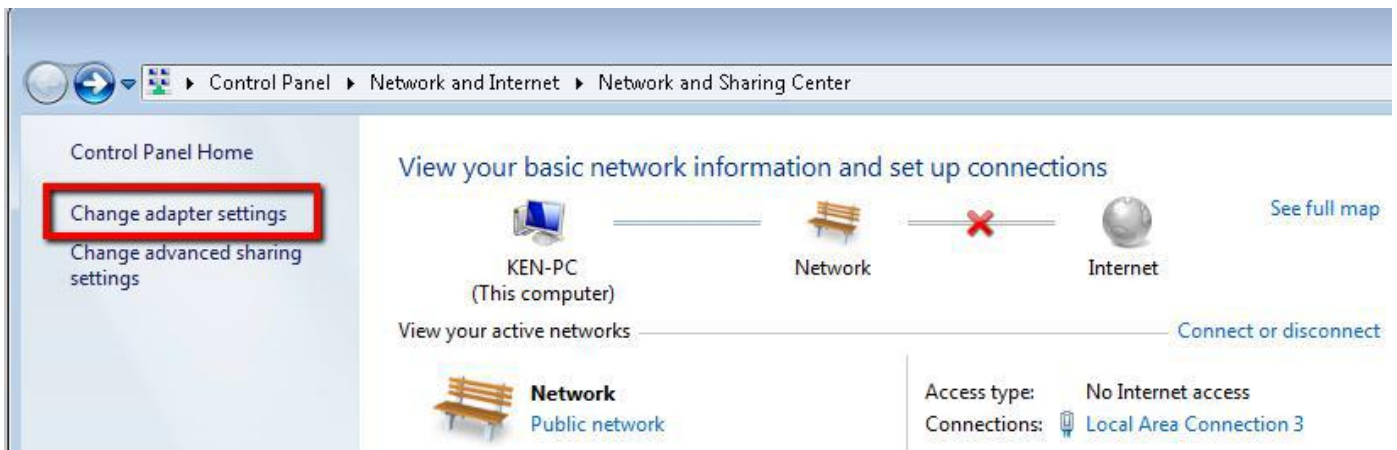


Figure 13 Adapter Setting

13. Choose the Ethernet as Figure 14.



Figure 14 PSP EBLE Ethernet

14. Right click "PSPEBLE" and select "Properties".

15. Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 15.

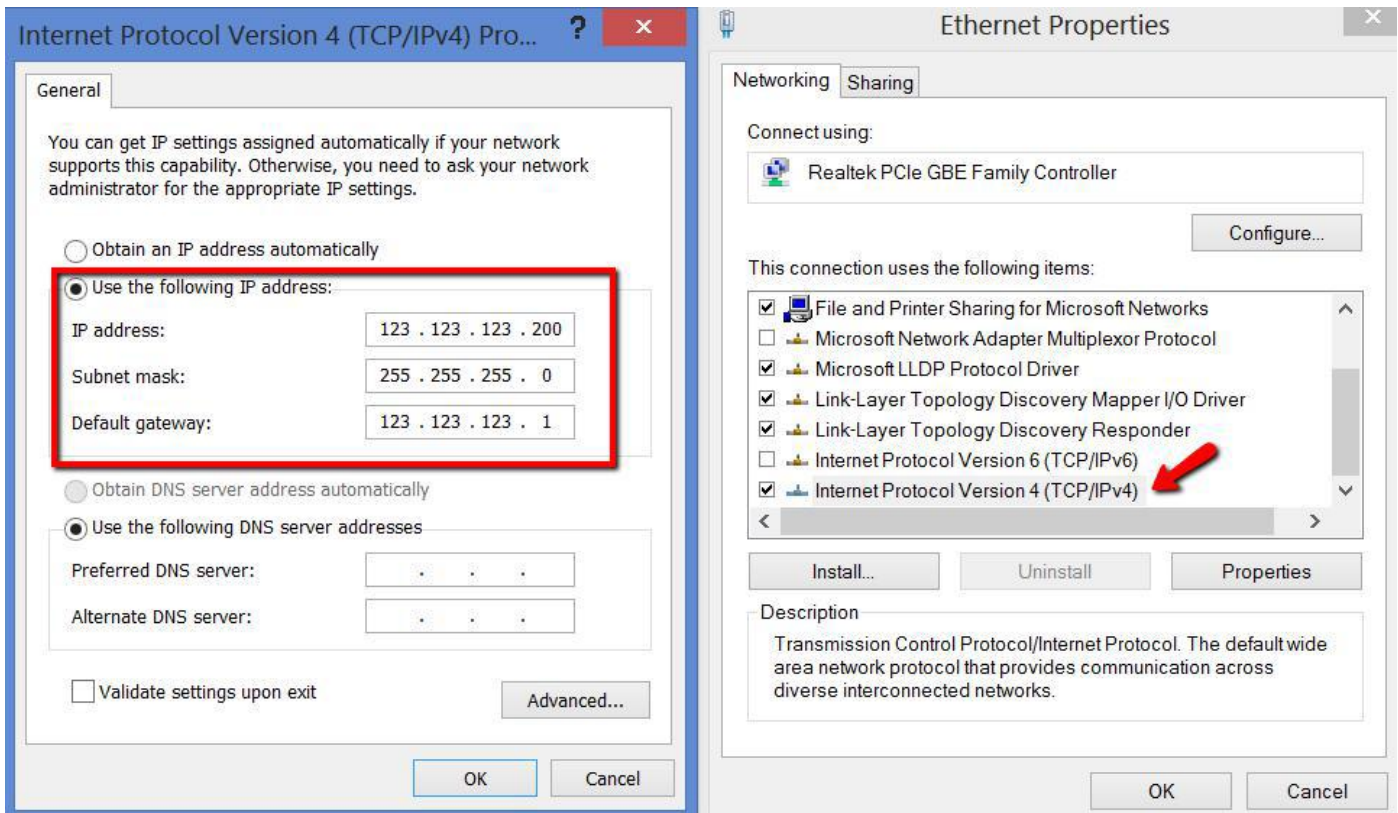


Figure 15 EBLE (PROJECTOR IP PORT) SETTING

16. Perform Ping Test at CMD by using OLD MACHINE IP & NEW MACHINE IP.

May refer to below table:

Type	IP ADDRESS (Front MODULE)
OLD MACHINE	192.168.0.100
NEW MACHINE	123.123.123.100

```

Command Prompt

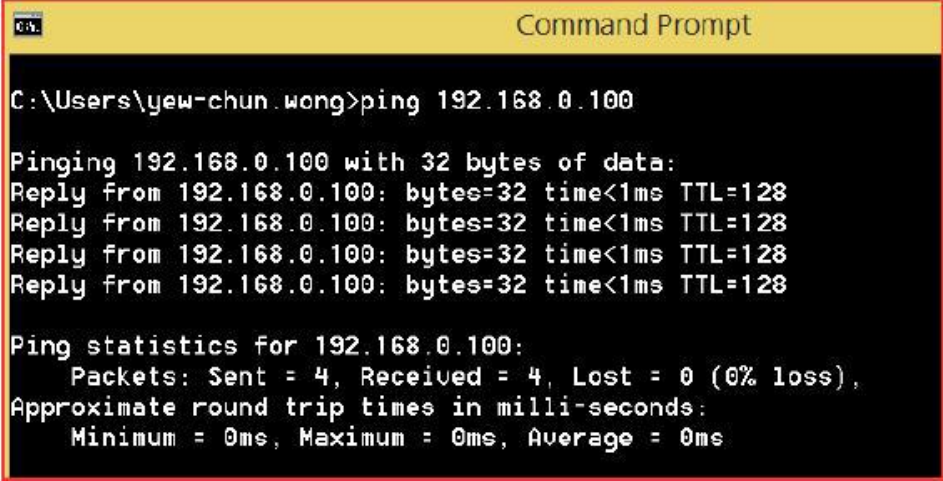
C:\Users\yew-chun.wong>ping 123.123.123.100

Pinging 123.123.123.100 with 32 bytes of data:
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128

Ping statistics for 123.123.123.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 16 New Front Module IP Setting (Reference Only)

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016



```

C:\Users\yew-chun.wong>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:
Reply from 192.168.0.100: bytes=32 time<1ms TTL=128
Reply from 192.168.0.100: bytes=32 time<1ms TTL=128
Reply from 192.168.0.100: bytes=32 time<1ms TTL=128
Reply from 192.168.0.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 17 Old Front Module IP Setting (Reference Only)

17. If IP Address “192.168.0.100” do not appear with return value as Figure 17, please refer Step 18-A to Step 18-R for Projector IP Configuration. Once Step 18-A to Step 18-R completed, may continue with Step 19.

If IP Address “123.123.123.100” is appear with return value as Figure 16 and may direct go to Step 19.

18. Perform Projector IP Configuration

- Go to Control Panel -> Network and Internet -> View Network and tasks.
- Choose “Change Adapter Settings”
- Choose the “PSP EBLE”
- Right click “PSP EBLE” and select properties.
- Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 18.

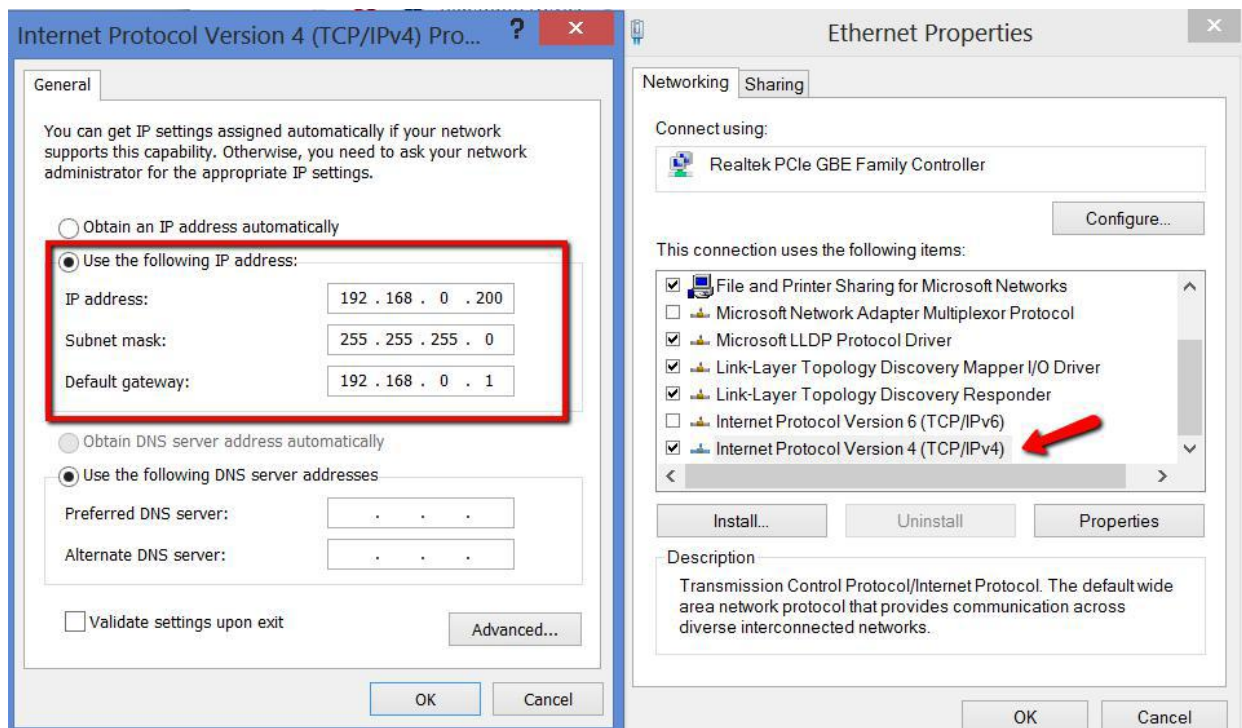


Figure 18 OLD IP SETTING

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

- f. Open up Internet Browser EG: Google Chrome, Internet Explorer or Mozilla Firefox.
- g. Type” <http://192.168.0.100>” as below Figure 19.

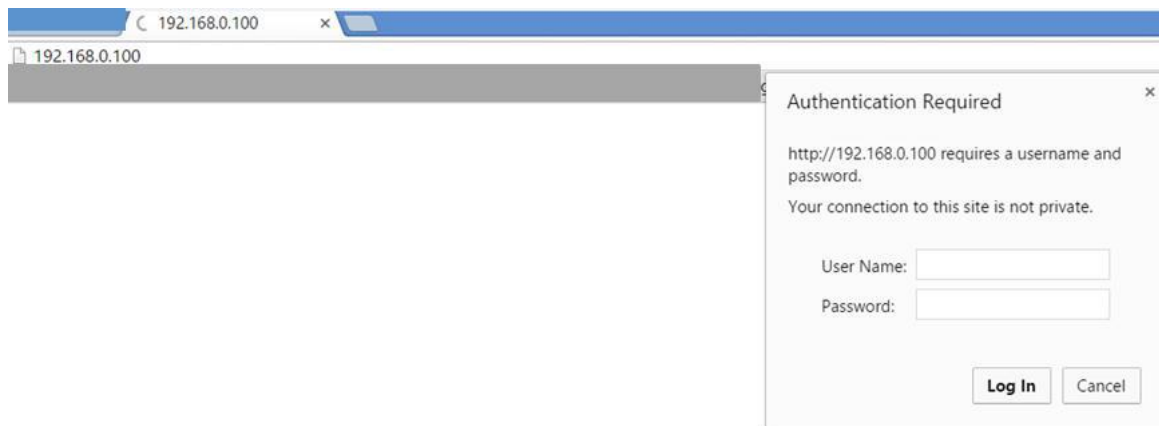


Figure 19 Front projector IP address

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

- h. Apply below information at the Authentication Dialog box. Please refer to below Figure 20.

User Name: Admin

Password : <Leave Blank>

Then, Press “Log In”

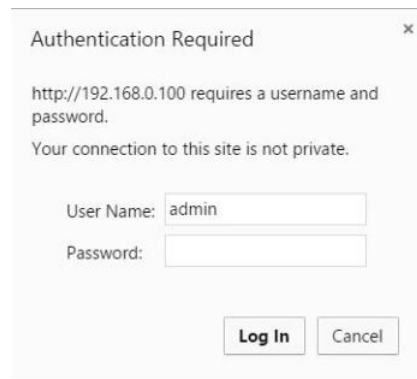


Figure 20 Authentication Dialog Box

- i. Network Setting dialog box will be displayed as below Figure 21.

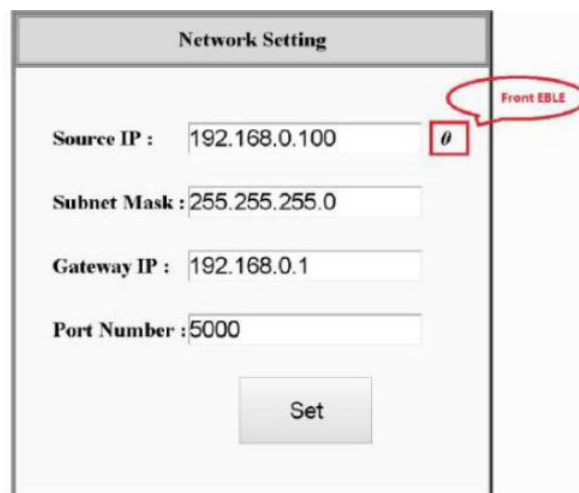


Figure 21 Front Projector Network Setting

- j. Perform IP Configuration by refer to below Figure 22.

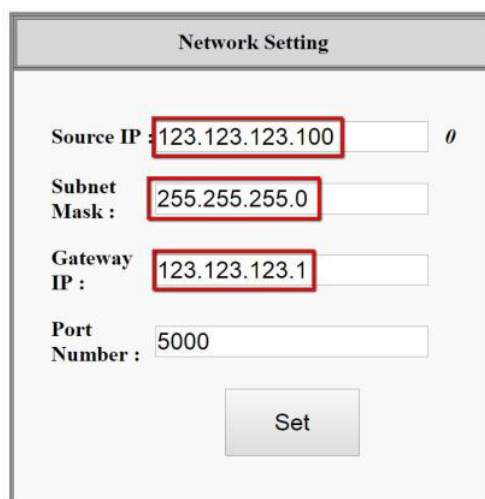


Figure 22 Front Projector New IP Setting

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

- k. Press “Set” and close Internet Browser
- l. Go to Control Panel -> Network and Internet -> View Network and tasks.
- m. Choose “Change Adapter Settings”
- n. Choose the “PSP EBLE”.
- o. Right click “PSP EBLE” and select “Properties”.
- p. Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 23.

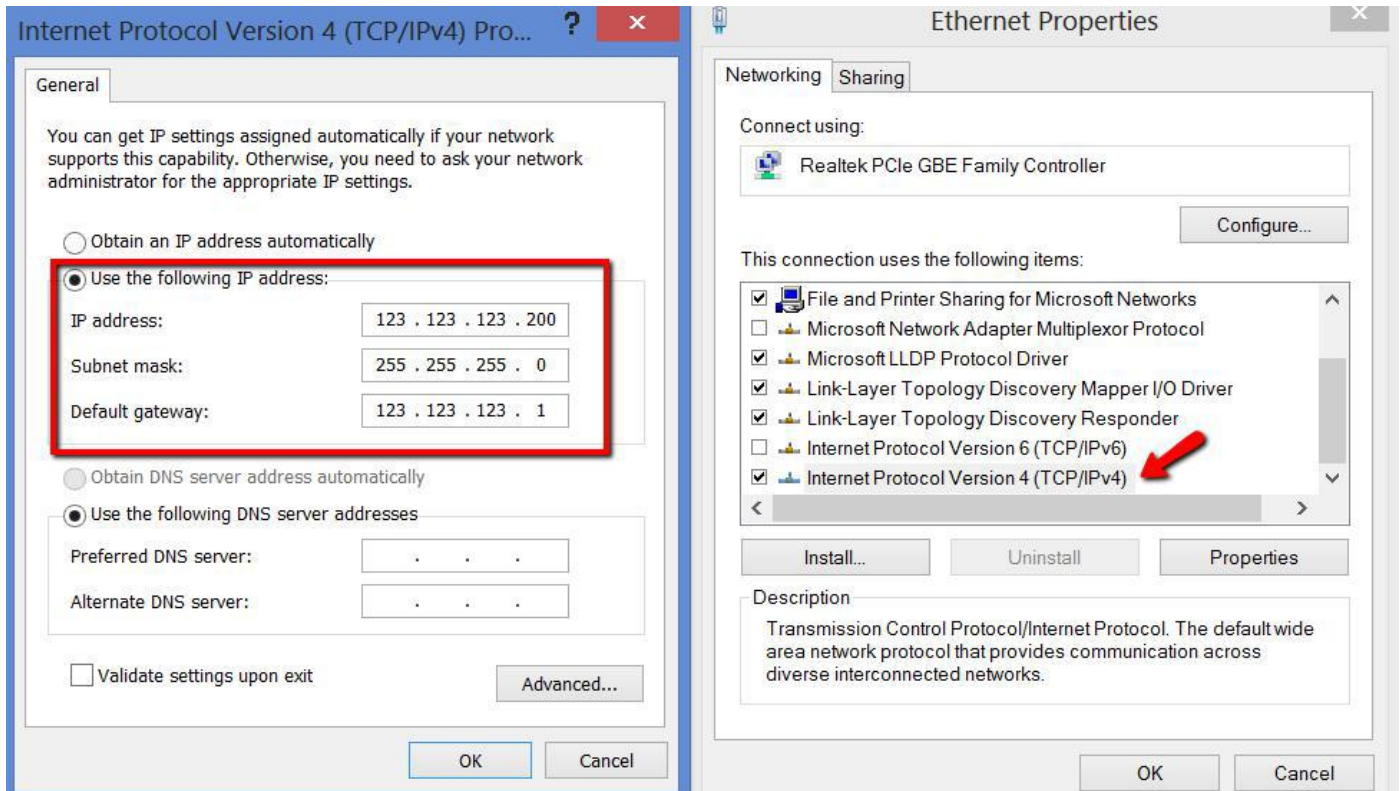
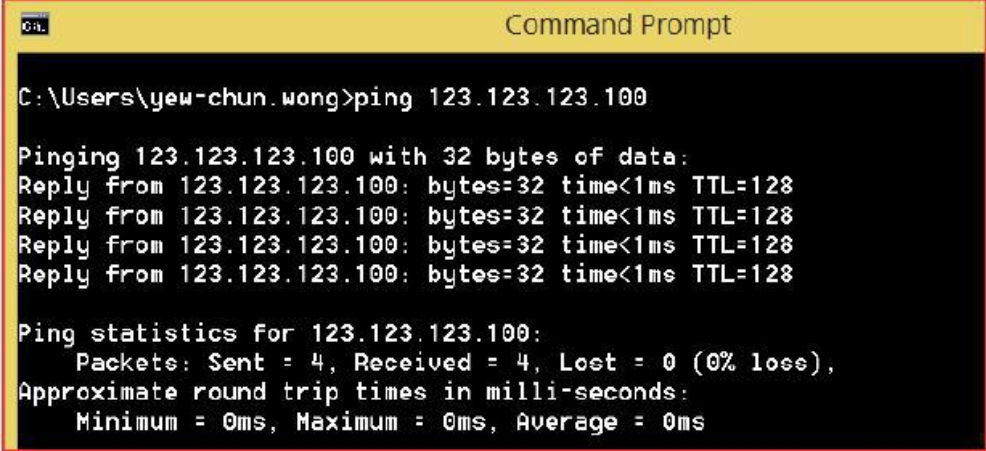


Figure 23 IP Setting

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

- q. Reseat Cat 5 cable at Projector Module.
- r. Perform Ping Test at CMD by type "ping 123.123.123.100"



```

C:\Users\yew-chun.wong>ping 123.123.123.100

Pinging 123.123.123.100 with 32 bytes of data:
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128
Reply from 123.123.123.100: bytes=32 time<1ms TTL=128

Ping statistics for 123.123.123.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 24 New Projector IP Address

19. Open Hardware Config by refer below path where depend Software Model :

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\XXLconfig
S2EX	C:\Program Files\AXI System\v810\5.0\s2exconfig

20. Change the key as below stated :

```

ethernetBasedLightEngineHostIpAddress=123.123.123.200
ethernetBasedLightEngineCurrentSettingMicroAmps=350
ethernetBasedLightEnginePeriod=20

ethernetBasedLightEngine0IpAddress=123.123.123.100
ethernetBasedLightEngine0Port=5000
  
```

21. Change key at Hardware Config as below Table.

System Type	Machine S/N	Adjustment Method	Hardware Config Key Changes
V810 S2EX	All	On Rail Adjustment Plate	pspModuleVersion=2 pspCalibrationVersion=3 pspSetting=2 opticalCameraModel=axiOpticalCameraType2

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

22. Open Software Config by refer to below path where depends on Software Model :

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\XXLconfig
S2EX	C:\Program Files\AXI System\v810\5.0\s2exconfig

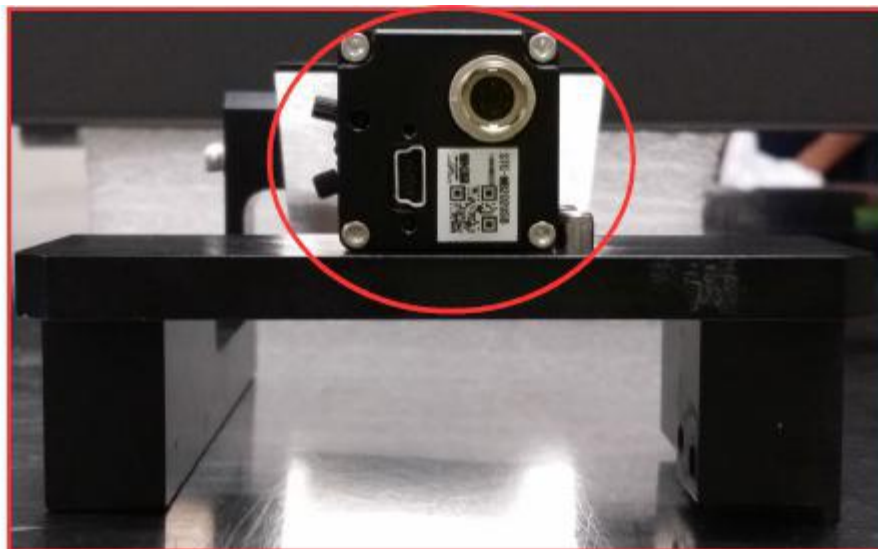
23. Change the key as below Figure 25:

```
# online or offline system
onlineWorkstation=true

# All hardware subsystems simulation
# Note: this parameter will override all of the individual hardware subsystems when set to
"true"
hardwareSimulation=false
```

Figure 25 Software Config

24. Identify Camera Orientation for Front PSP 2 Module as below Figure 26.



Front Camera Position Orientation

Figure 26 Front PSP 2 Module (Camera Orientation)

25. Open StCamWare at Window Desktop.as below Figure 27.



Figure 27 StCamsWare

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

26. Optical Camera Live View Dialog box would be illustrated as below Figure 28.

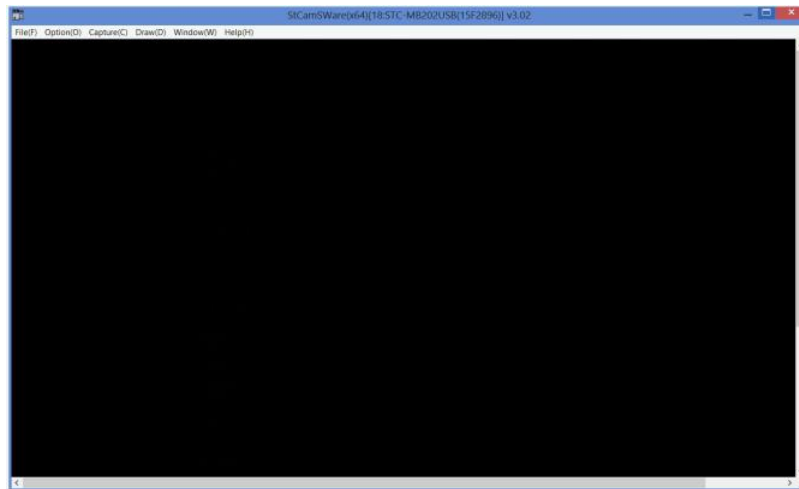


Figure 28 Optical Camera Live View

27. Go to “Option” and select Rename Camera (R) as below Figure 29.

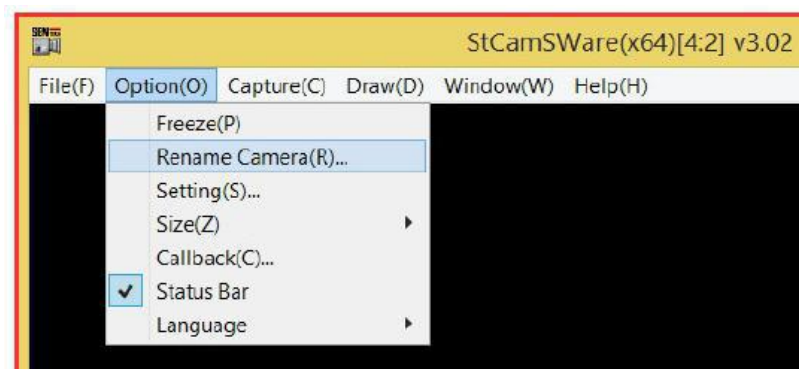


Figure 29 Optical Camera Live View

28. Camera Name Dialog box would be illustrated as below Figure 30.

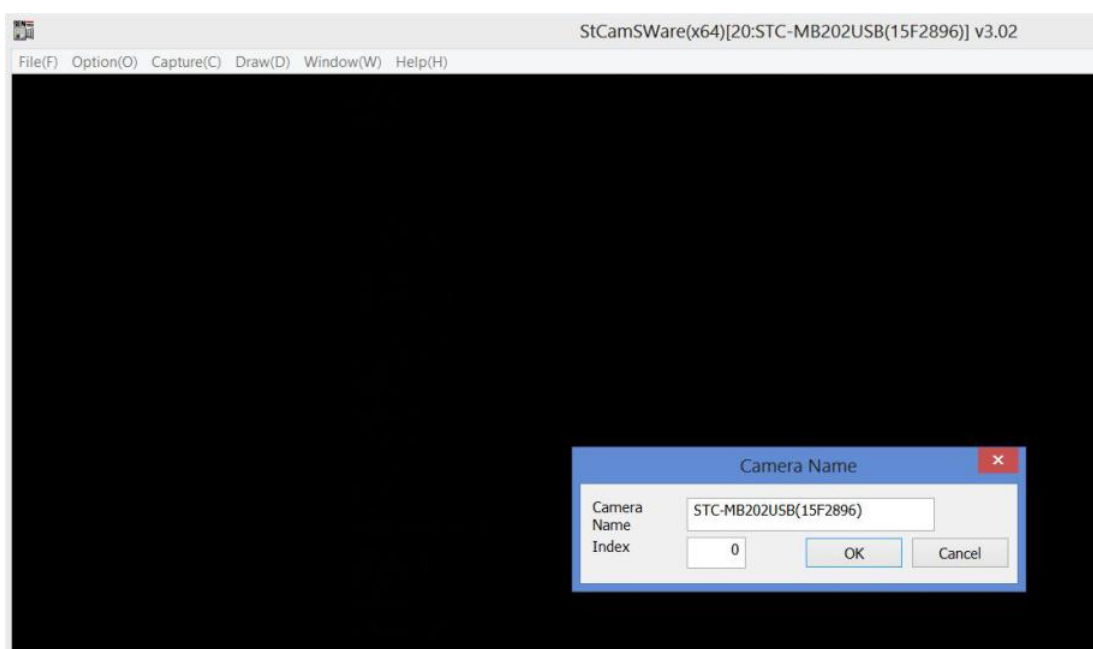


Figure 30 Camera Name Dialog Box

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

29. Configure Front PSP 2 Camera Module by refer to below Figure 31 and Click “OK”.

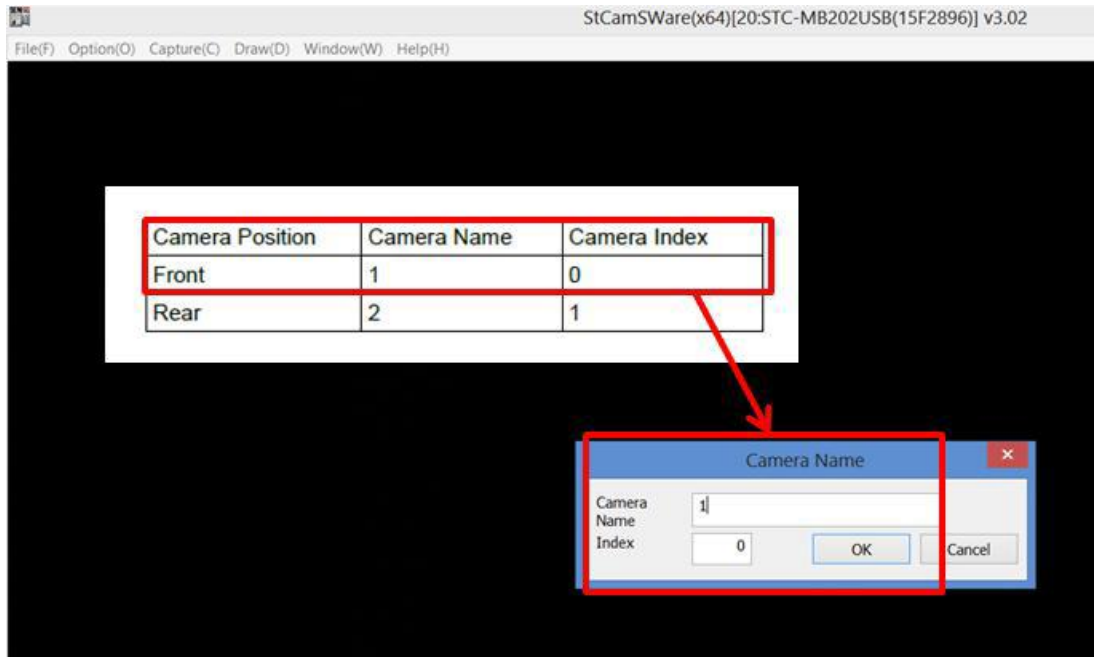


Figure 31 Camera Name Dialog box

30. Open V810 Software & go to Services-> Subsystem Status& Control-> Ethernet-Based Light Engine and press Initialized All EBLEs as below Figure 32.

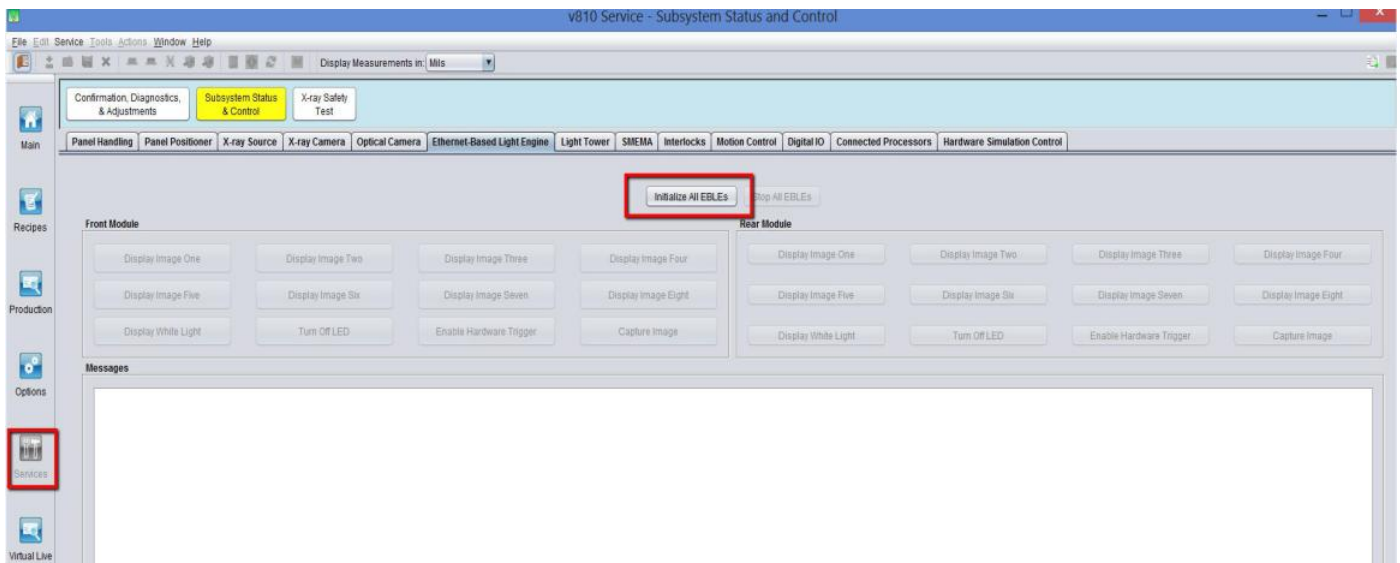


Figure 32 EBLE GUI

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

31. At Front Module Column, please click “Display White Light” as below Figure 33.

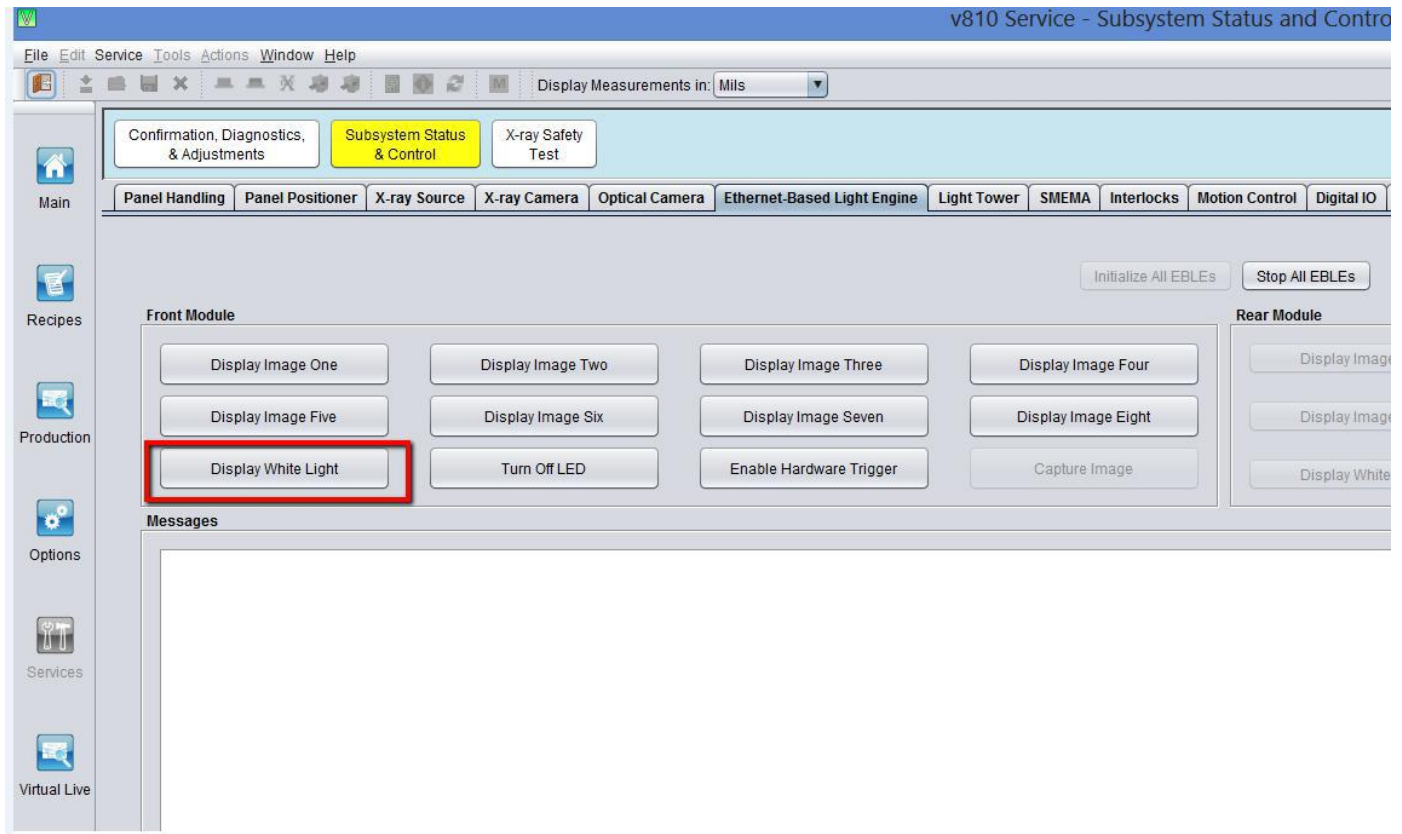


Figure 33 Front Module Column

32. At StCamsWare , Go to Option and Select Setting as below Figure 34.

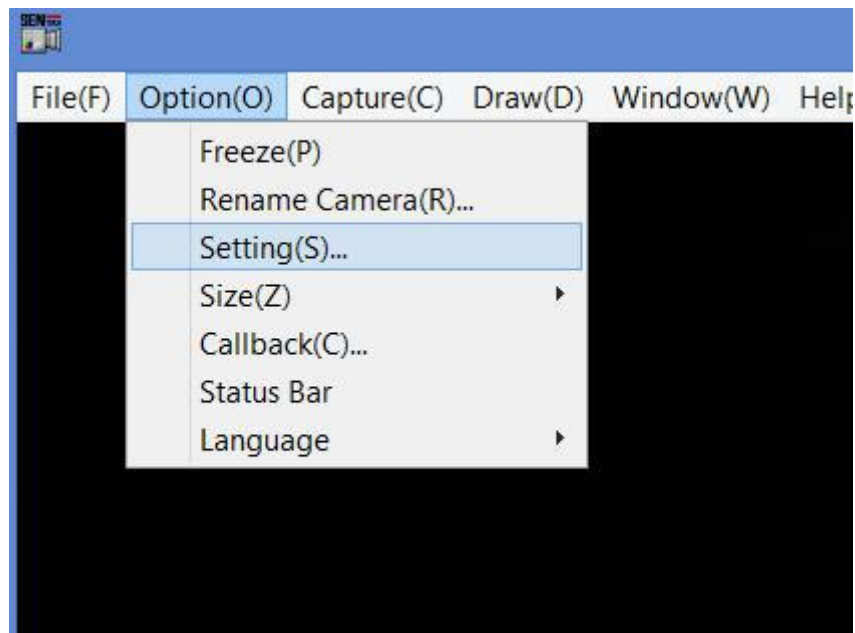


Figure 34 StCamsWare GUI

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

33. Select “OFF (Normal)” at Mirror Image Column as below Figure 35.

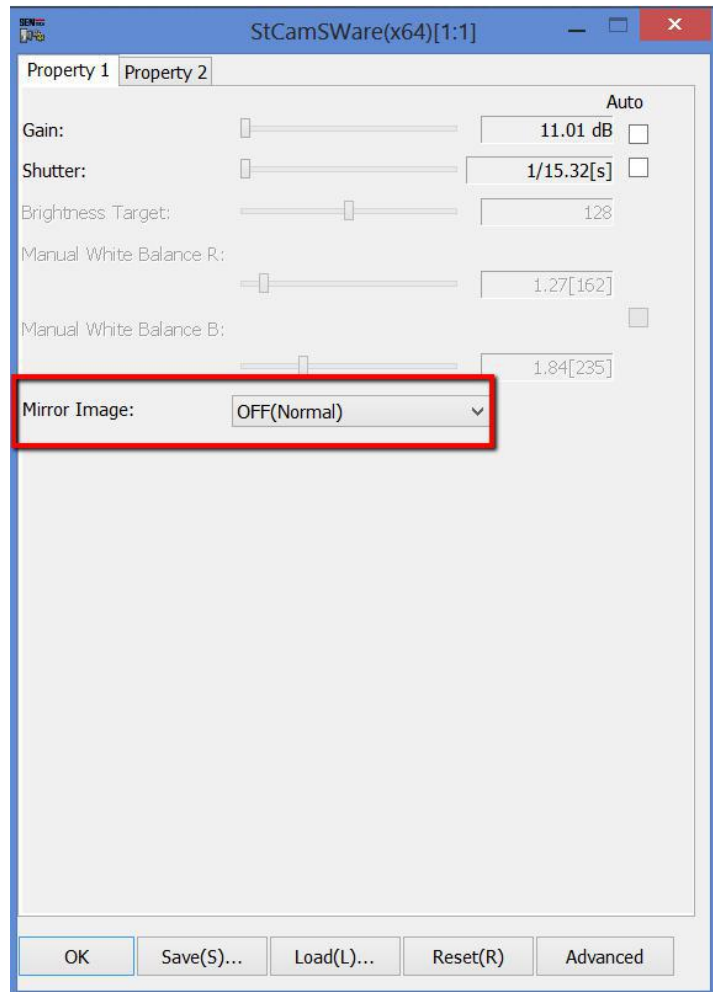


Figure 35 StCamsWare GUI

34. Loosen 2 x Screws and extend the module towards Front Direction as below Figure 36.

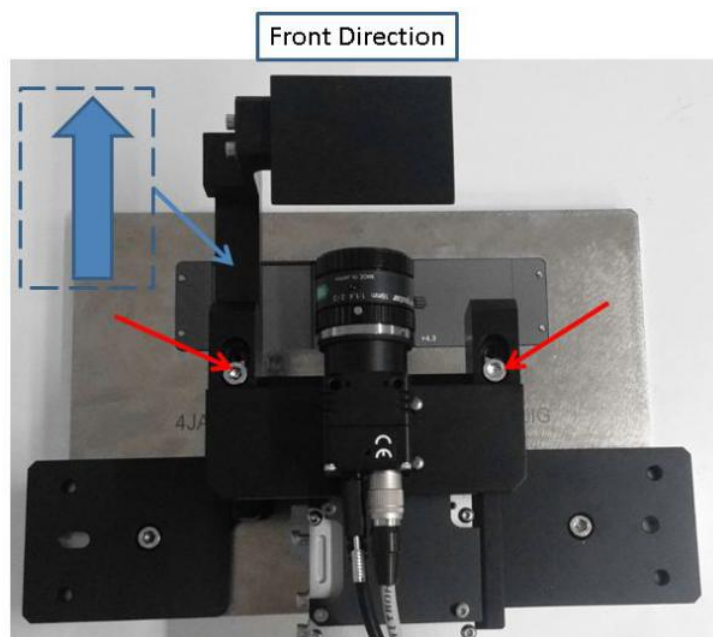


Figure 36 Module Extend towards Front Direction

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

35. Loosen 2 x Side Screws of Projector Module as below Figure 37.

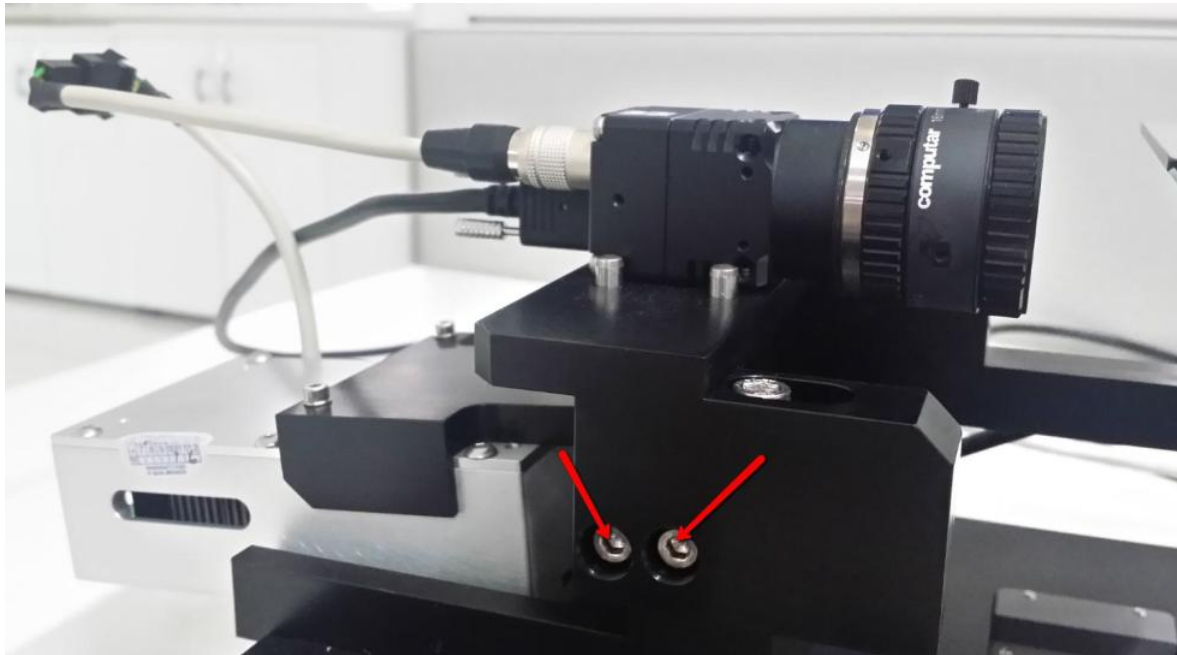


Figure 37 Projector Module Side Screws

36. Adjust the Projector angle until white light projection is fully covered on the calibration plate as below Figure 38.

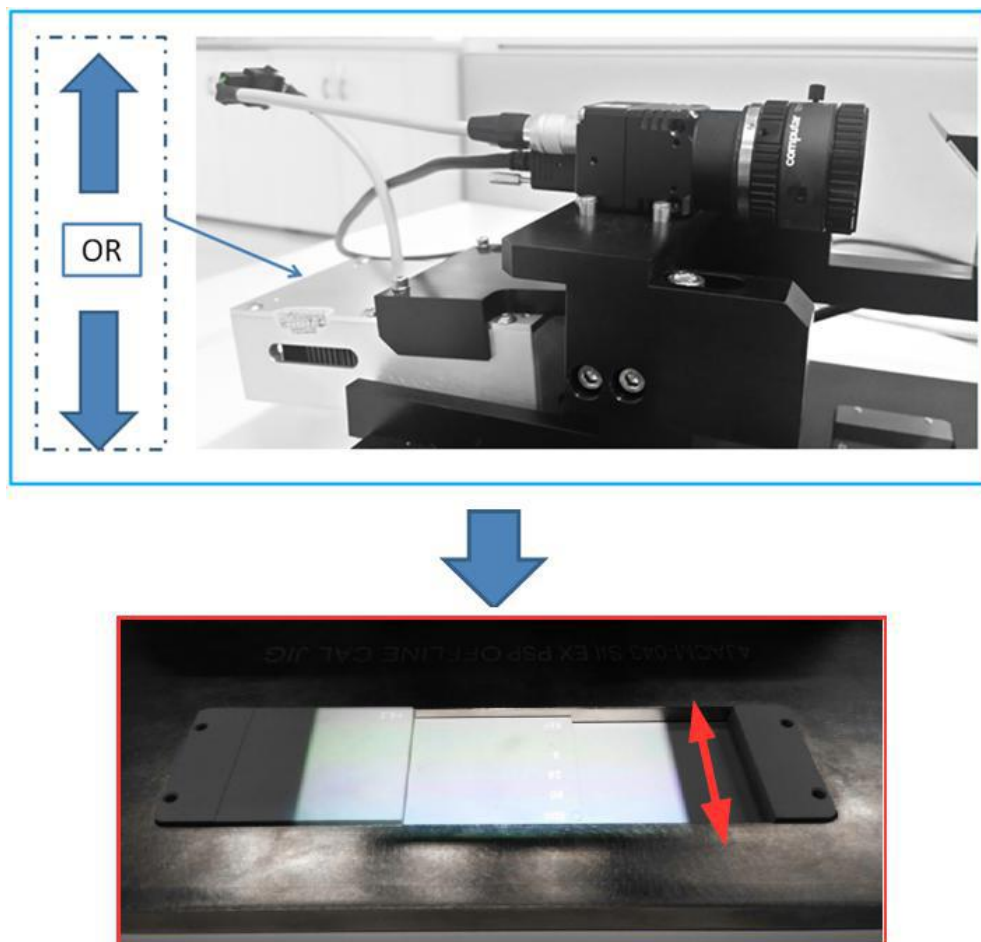


Figure 38 White Light Projection Coverage Area

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

37. Go to StCamSware, go to Option -> Size, Select “x 1” as below Figure 39.

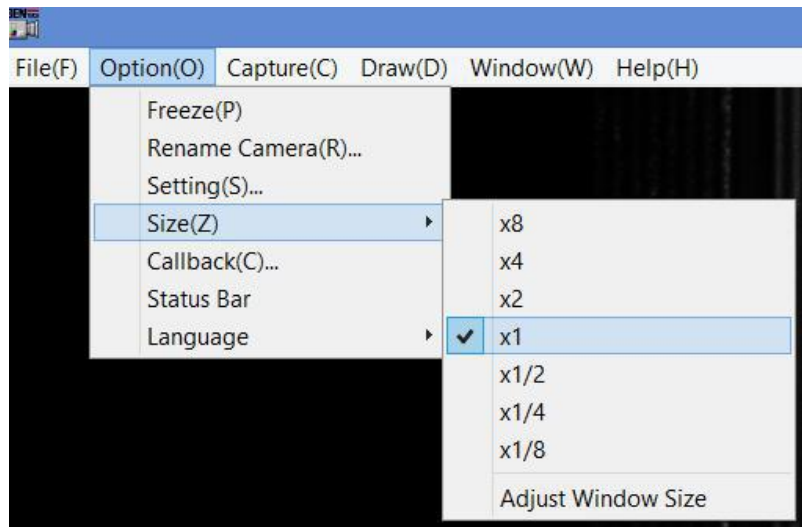
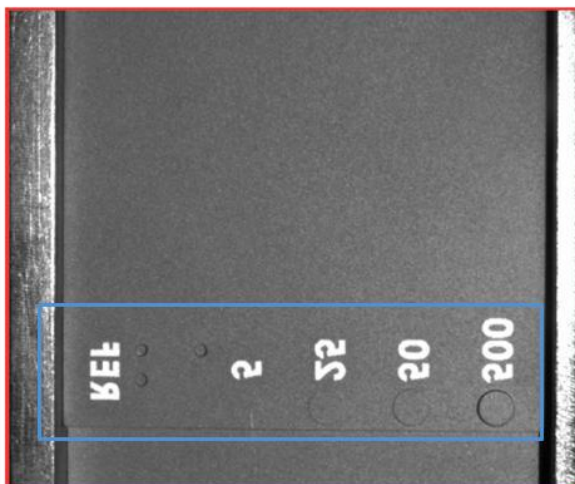
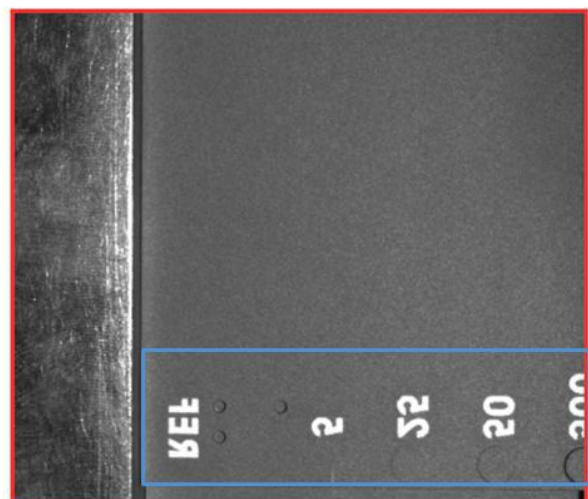


Figure 39 StcamSware GUI

38. Open StCamsWare and verify the white light projection image is within the desired coverage area as below Figure 40.



Full Coverage



Insufficient Coverage parameter

Figure 40 White Light Coverage area

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

39. Next, Go to StCamSware, go to Option -> Size, Select “x ½” as below Figure 41.

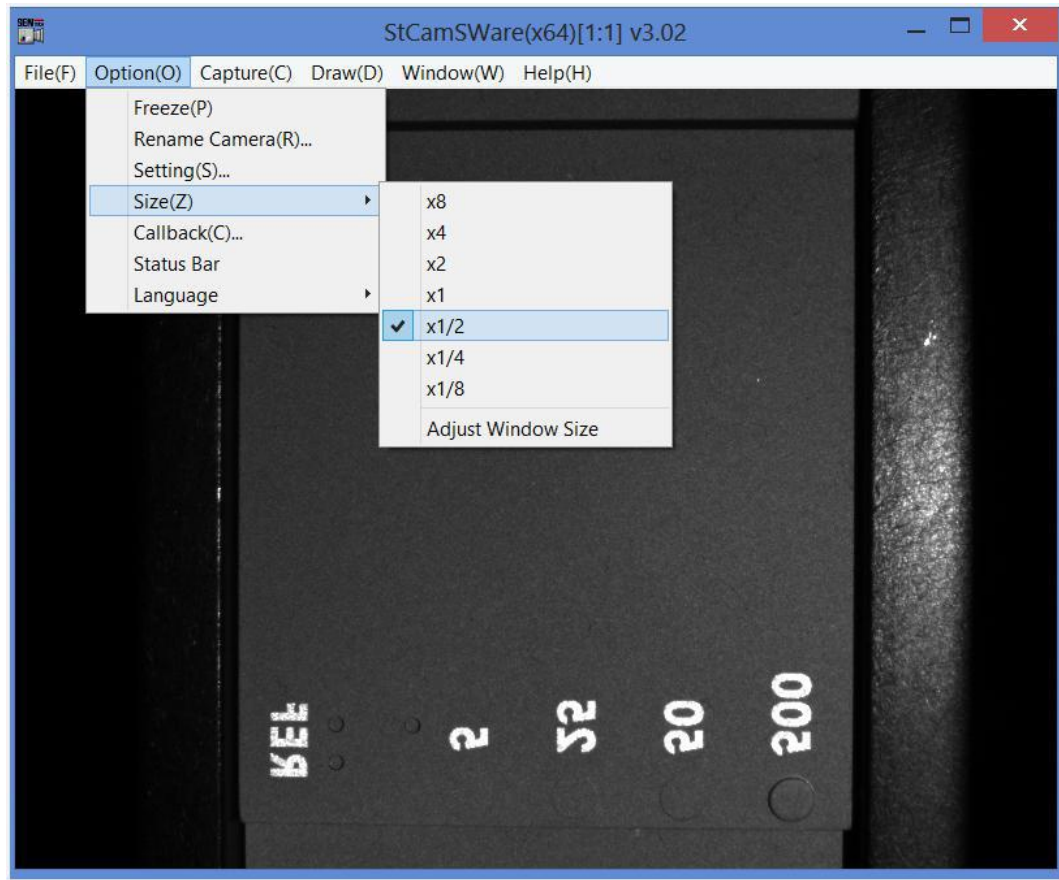


Figure 41 StCamSware Frame Size Setting

40. At EBLE tab, Trigger “Display White Light” at Front Module Column.

Remarks: If white light had been triggered previously, may ignore this Step.

41. Loosen the “Tiny knob” at the optical camera as below Figure 42.



Figure 42 Optical camera

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

42. Adjust Optical Camera Focus by turning the dial of optical camera as below Figure 43.



Figure 43 Optical Camera Dial Turning

43. Verify wording on all plane in focus. (REF plane, 5 plane, 25 Plane, 50 plane, 500 plane) as below Figure 44. Repeat Step 42 to 43 until all are plane in focus. Tighten the dial of optical camera once completed.

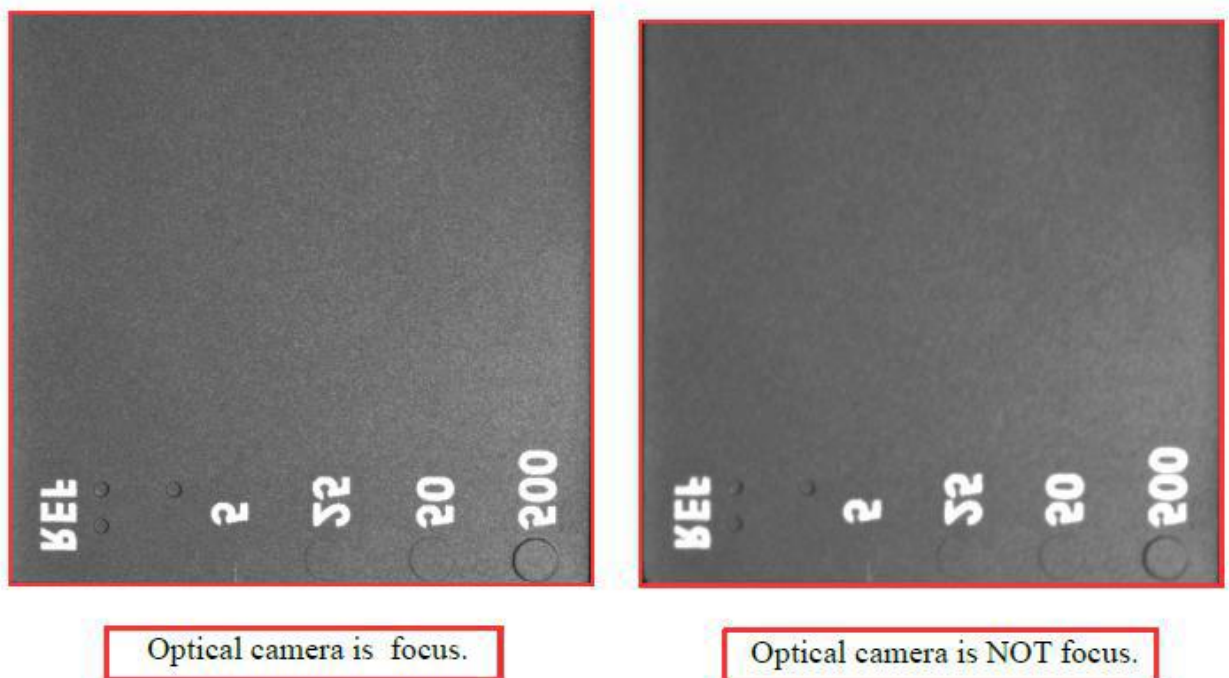


Figure 44 All Plane Focusing

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

44. Next, Go to EBLE tab at V810 Software and Select “Display Image Two” as below Figure 45.

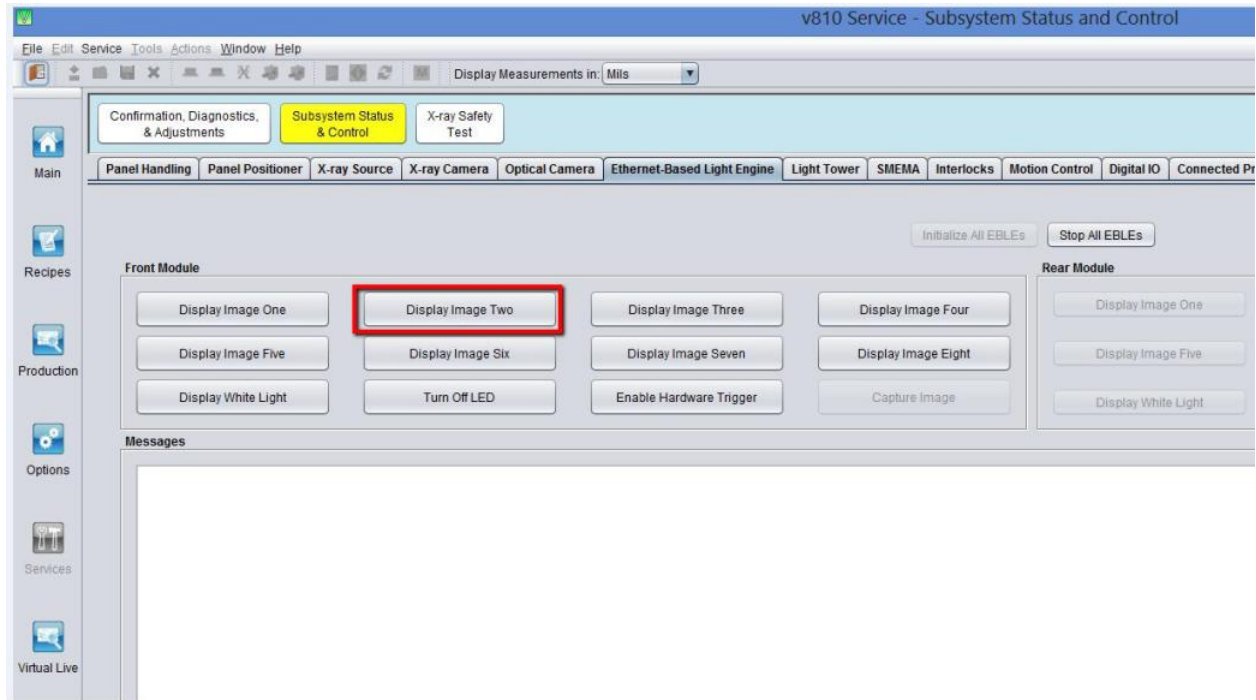


Figure 45 EBLE GUI

45. Fringe pattern would illustrate at StCamSware GUI as below Figure 46.



Figure 46 Fringe Pattern

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

46. Go to StCamSware GUI ->Option and Select Size (Z) “x1” as below Figure 47.

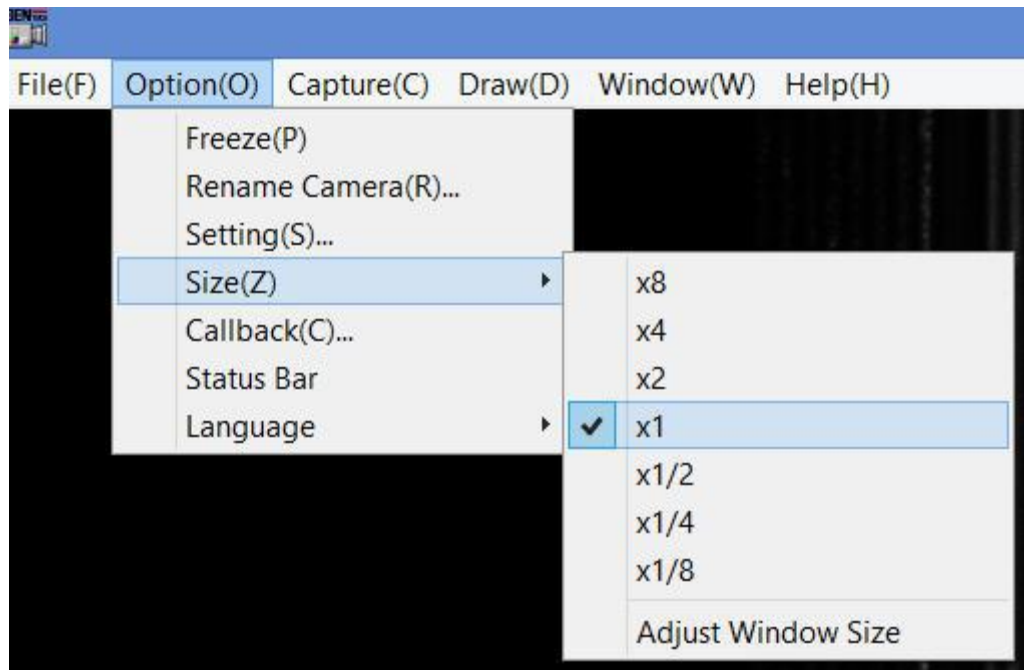


Figure 47 StCamSware GUI

47. Loosen 2 X Screws at EBLE Projector as below Figure 48.

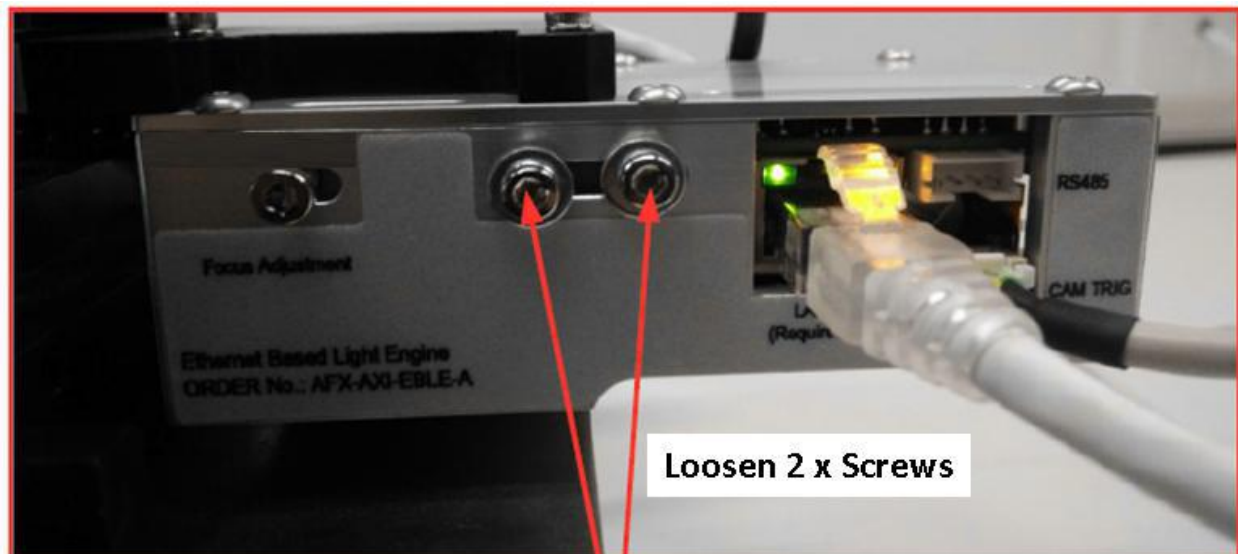


Figure 48 EBLE Projector

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

48. Adjust the Focus Adjustment until “Clear Fringe Pattern” is obtained as Figure 49. Then, Tighten 2 x Screws.

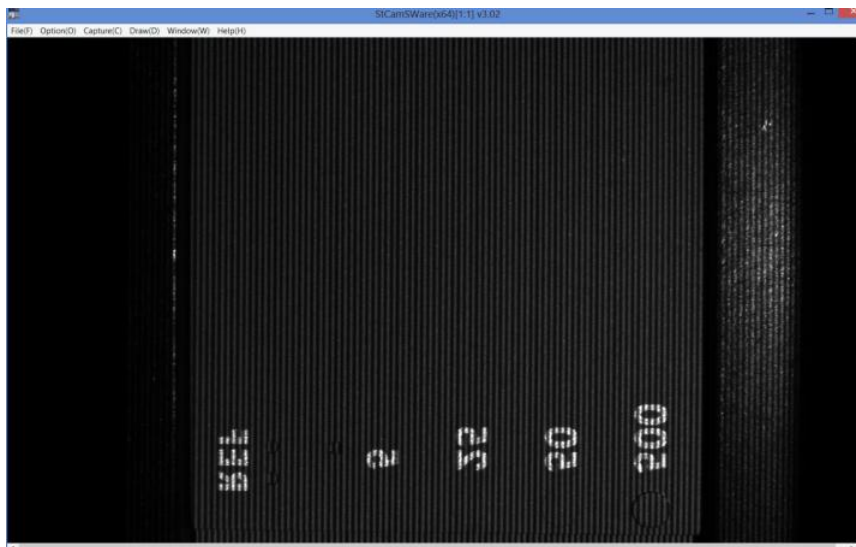
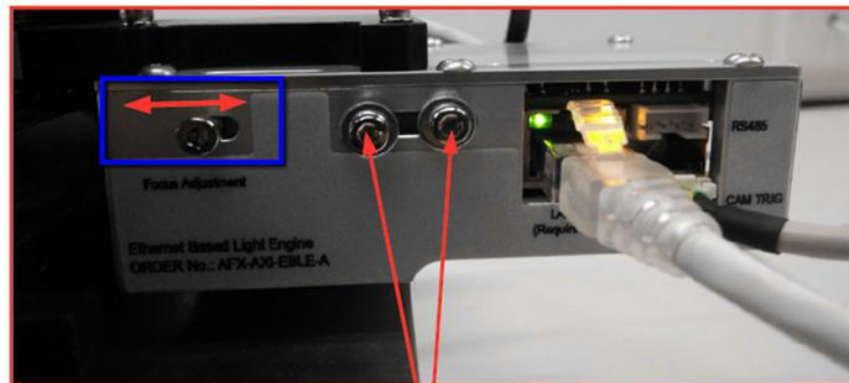


Figure 49 Clear Fringe Pattern

49. Then, Go to StCamSware GUI ->Option and Select Size (Z) “x 8” as below Figure 50.

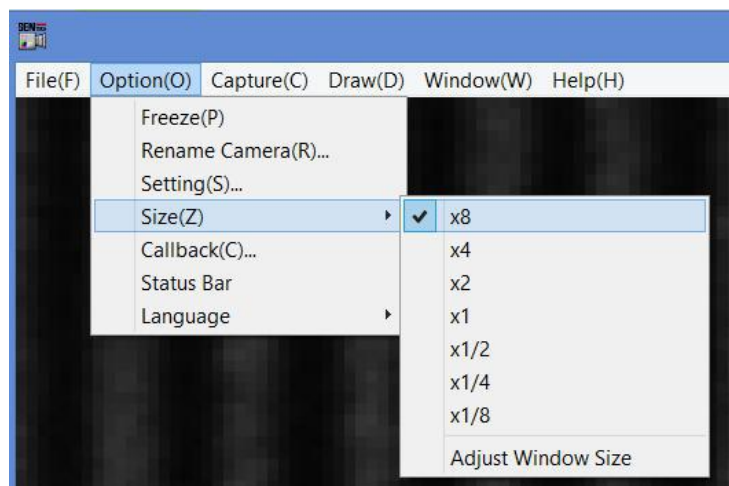


Figure 50 StcamSware GUI

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

50. Verify the Fringe pattern image sharpness are approximately similar to Image A (Figure 51)

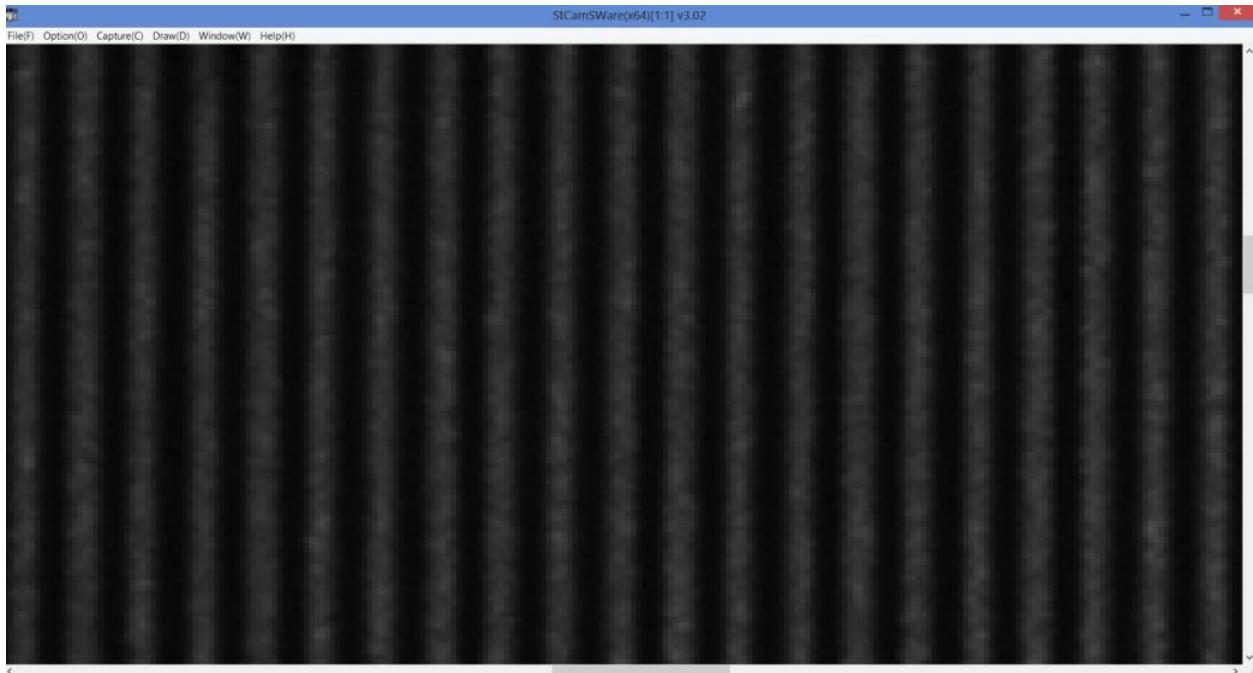


Figure 51 Image A

- s. If Image A is obtained during the verification, may direct proceed to Step 51.
- t. If Image B is obtained during the verification, please re-adjust the “Focus adjustment” of projector by refer to “Size 8” at Stcamsware GUI until Image A (Figure 51) is obtained. Once completed, please re-verify the sharpness image at “Size 1” at Stcamsware GUI as Figure 49.

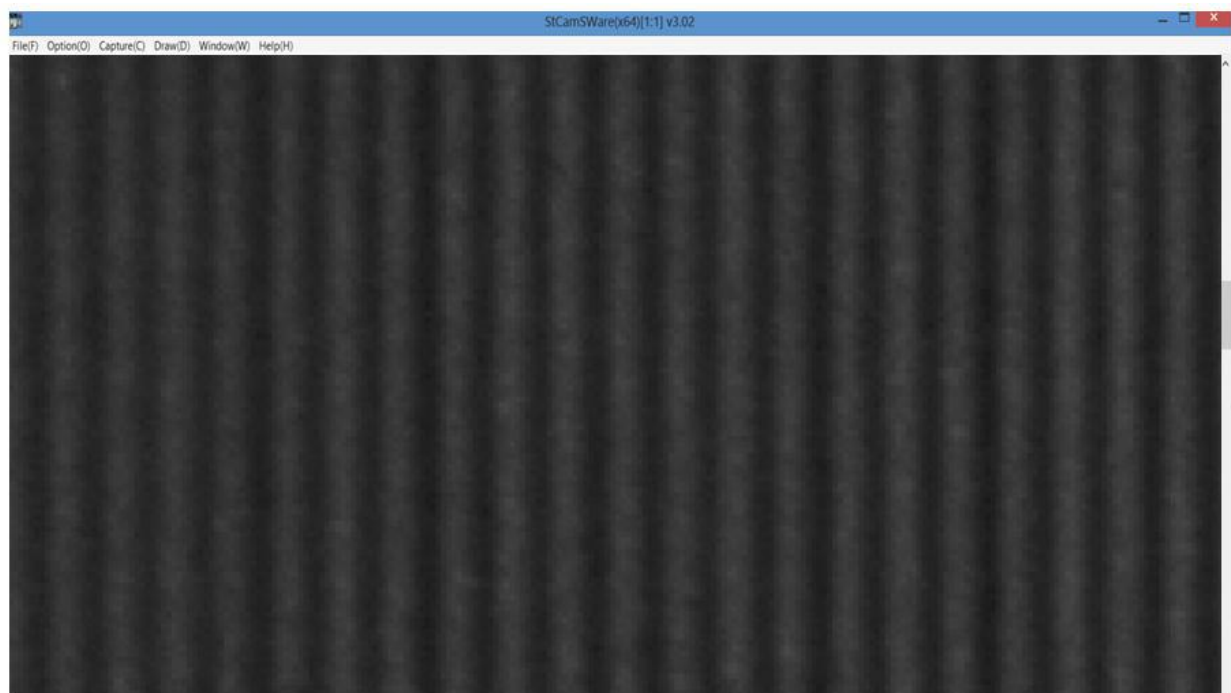


Figure 52 Image B

51. Next, Close the StcamSware Software.

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

52. Obtain “Aperture_StCamWare” from AXI HW Support Team. Figure 52 as below.



Figure 52 Aperture_StCamWare

53. Double Click “Aperture_StCamWare” and 3 file will be displayed as below Figure 53.

 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe_S2EX	7/31/2015 5:30 PM	EXE_S2EX File	192 KB
 StTrgDisplayVCS2005.exe_StdXXL	7/27/2015 10:56 A...	EXE_STDXXL File	192 KB

Figure 53 Aperture_StCamWare File Type

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

54. Rename the “**StTrgDisplayVCS2005.exe_S2EX**”to“**StTrgDisplayVCS2005.exe**” as below Figure 54.

Name	Date modified	Type	Size
 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe_S2EX	7/31/2015 5:30 PM	EXE_S2EX File	192 KB
 StTrgDisplayVCS2005.exe_StdXXL	7/27/2015 10:56 A...	EXE_STDXXL File	192 KB



 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe	7/31/2015 5:30 PM	Application	192 KB
 StTrgDisplayVCS2005.exe_StdXXL	7/27/2015 10:56 A...	EXE_STDXXL File	192 KB

Figure 54 Exe File rename

55. Then, Go to EBLE tab at V810 Software and select “Display White Light “.

56. Double click the new rename file (**StTrgDisplayVCS2005.exe**) where under Aperture_StCamWare Folder.

57. A Dialog box with live View will be illustrated as below Figure 55.

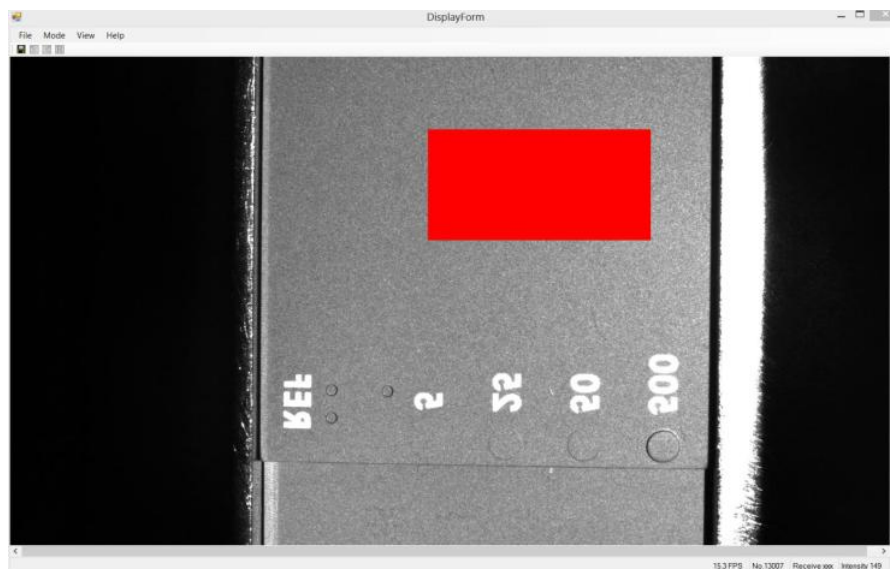


Figure 55 Display Form Dialog Box

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

58. Loosen the “Tiny Knob” at Optical Camera as below Figure 56.



Figure 56 Tiny Knob at OpticalCamera

59. Turning the dial for Optical Camera Aperture as below Figure 57.



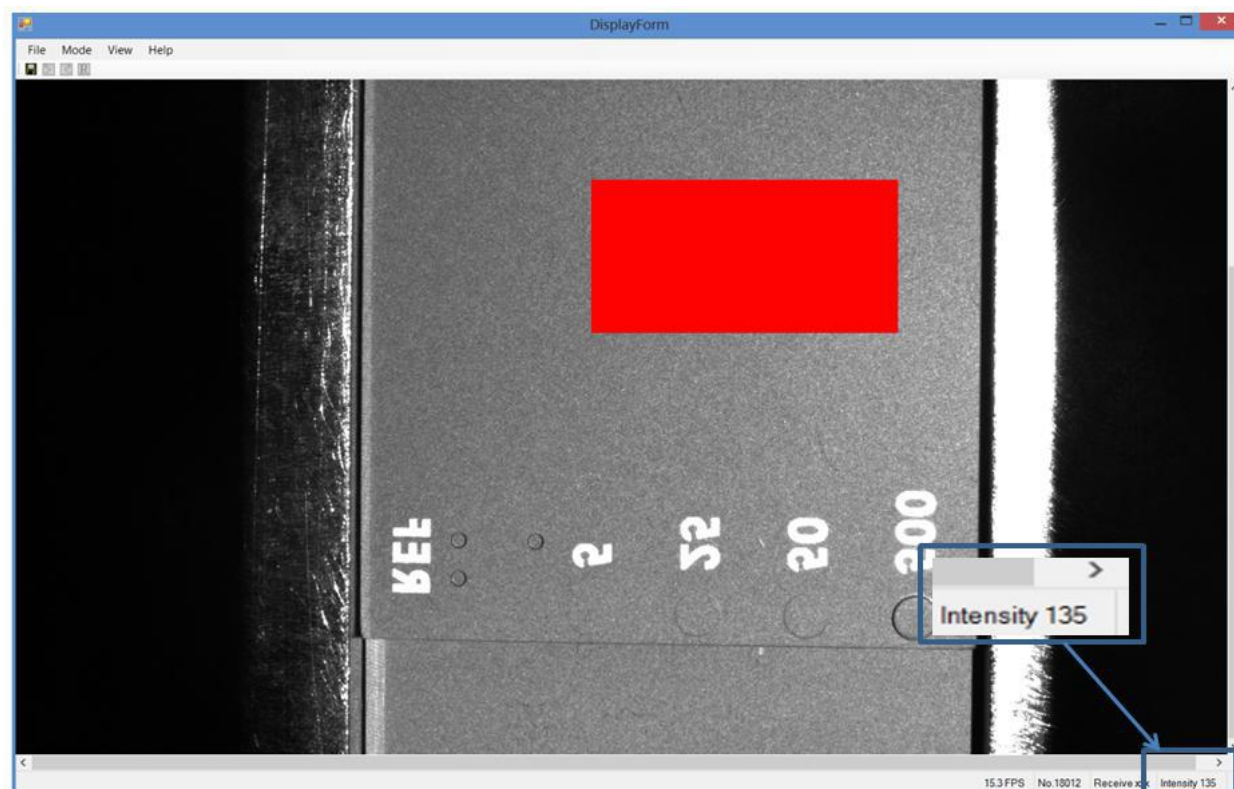
Figure 57 Aperture Dial Turning

60. Stop the turning when Intensity is within below range as below Figure 58. Once complete, tighten the “Tiny Knob” where previous mentioned at Step 58.

Intensity range	132-138
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Poor Adjustment



Well Adjustment

Figure 58 Intensity range

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX PSP Front Module Offline Adjustment Procedure (FOR EQUIPMENT)	Effective Date	2 Oct 2016

61. Click “ Stop All EBLE” at EBLE Tab (V810 Software)
62. Close the “Display Form Dialog Box” as Figure 59.
63. Under “Aperture_StCamWare” Folder and rename the
“StTrgDisplayVCS2005.exe”to“StTrgDisplayVCS2005.exe_S2EX” as below Figure 59.

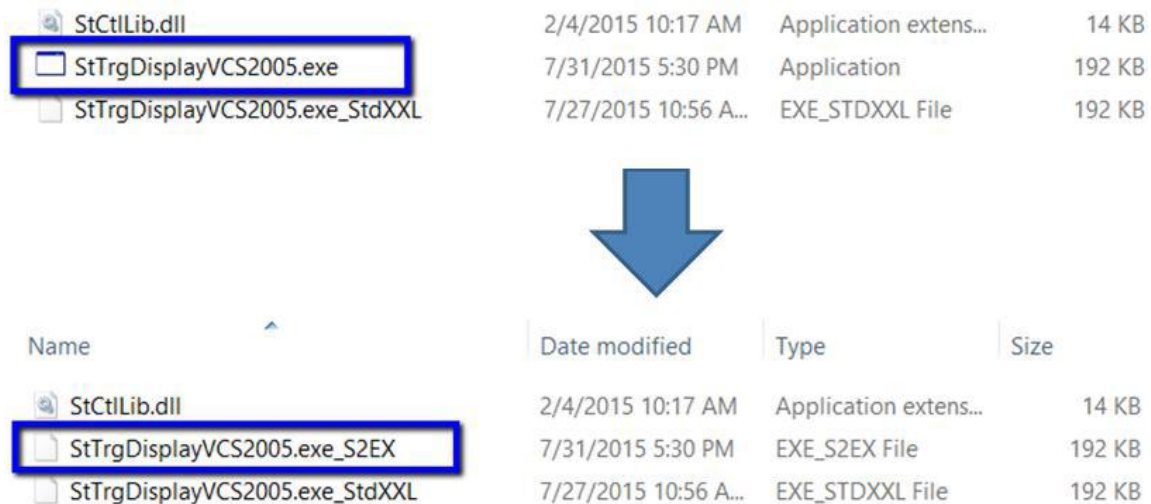


Figure 59 Exe File rename

64. Unplug 1 x Blue Cat 5 Cable from PSP 2 Module
65. Revert to Origin Cat 5 Cable at POE Switch.
66. Remove the Optical Camera “Area Scan” Cable from PSP 2 Module and master controller.
67. Remove the entire Module and Offline Adjustment Jig from the machine.
68. Exit V810 Software.
69. Go to hardware config and change as below table.

System Type	Machine S/N	Adjustment Method	Hardware Config Key Changes
V810 S2EX	All	On Rail Adjustment Plate	pspSetting=1

70. Re-open V810 Software.